

EXTENSION 1

Stage 6

Calculus (Ext1), C3 Applications of Calculus (Y12), Differential Equations (Ext1)

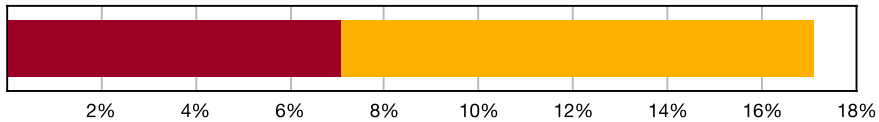
Equations and Slope Fields

Teacher: SHS Maths

Exam Equivalent Time: 96 minutes (based on allocation of 1.5 minutes per mark)



C3 Applications of Calculus



Further Area and Solids of Revolution
Differential Equations

*SmarterMaths analytics based on the average contribution to Extension 1 exams since 2020.

HISTORICAL CONTRIBUTION

- C3 Application of Calculus* is the 880-pound Silverback of the Extension 1 course, contributing a huge 17.1% per new syllabus Ext1 exam since the new course began in 2020.
- This topic has been split into two sub-categories for analysis purposes which are: *1-Further Areas and Solids of Revolution (7.1%)* and *2-Differential Equations (10.0%)*.
- This analysis looks at a sub-topic of *Differential Equations* → *Equations and Slope Fields (5.7%)*.

HSC ANALYSIS - What to expect and common pitfalls

- Equations and Slope Fields* have appeared in multiple questions in all but one new syllabus exam, making this an important focus area for any revision.
- Solving differential equations where the derivative is $f(y)$ or $f(x, y)$ is a core competency that has been examined in 3 of the last 4 exams (absent in 2023), including 2 questions worth a total 5 marks in 2020! A key revision area.
- 2022 Ext1 14a produced an outlier mean mark within this area of 43% and deserves specific revision attention.
- Slope fields* is a "highly examinable" topic area that has also been examined in each new syllabus exam (note the 2024 example appears later in *Applications of DE*). 2021 Q12a asked students to sketch a solution curve and caused major problems with a mean mark of 18%.

Questions

1. Calculus, EXT1 C3 2021 HSC 4 MC

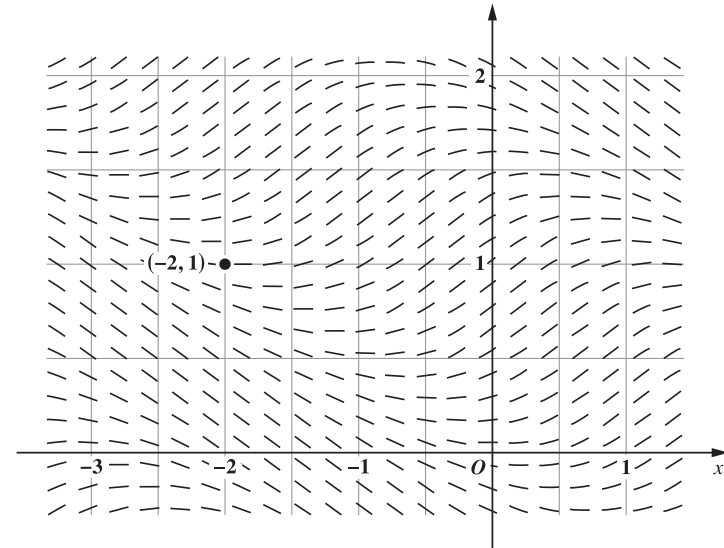
Consider the differential equation $\frac{dy}{dx} = \frac{x}{y}$.

Which of the following equations best represents this relationship between x and y ?

- A. $y^2 = x^2 + c$
- B. $y^2 = \frac{x^2}{2} + c$
- C. $y = x \ln |y| + c$
- D. $y = \frac{x^2}{2} \ln |y| + c$

2. Calculus, EXT1 C3 2023 HSC 3 MC

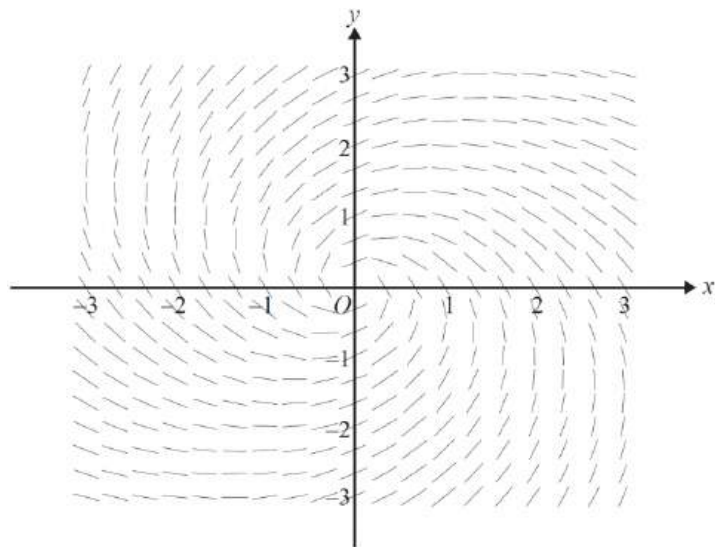
The diagram shows the direction field of a differential equation. A particular solution to the differential equation passes through $(-2, 1)$.



Where does the solution that passes through $(-2, 1)$ cross the y -axis?

- A. $y = 1.12$
- B. $y = 1.34$
- C. $y = 1.56$
- D. $y = 1.78$

3. Calculus, EXT1 C3 2011 SPEC2 17 MC



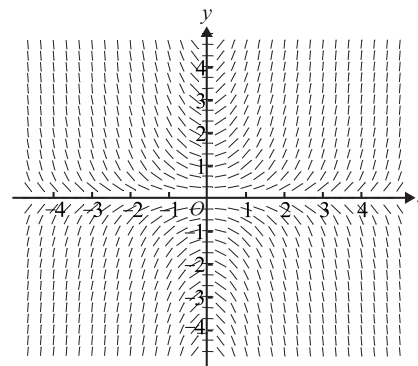
The differential equation which best represents the above direction field is

- A. $\frac{dy}{dx} = \frac{y - 2x}{2y + x}$
- B. $\frac{dy}{dx} = \frac{2x - y}{y - 2x}$
- C. $\frac{dy}{dx} = \frac{2y - x}{y + 2x}$
- D. $\frac{dy}{dx} = \frac{y - 2x}{2y - x}$
- E. $\frac{dy}{dx} = \frac{x - 2y}{2y + x}$

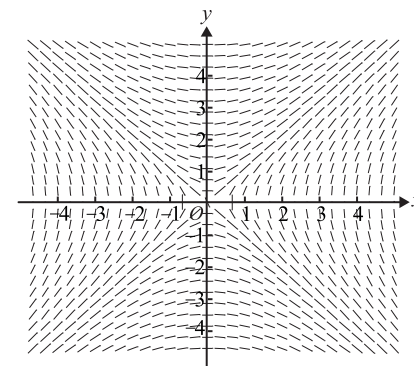
4. Calculus, EXT1 C3 2012 SPEC2 10 MC

The diagram that best represents the direction field of the differential equation $\frac{dy}{dx} = xy$ is

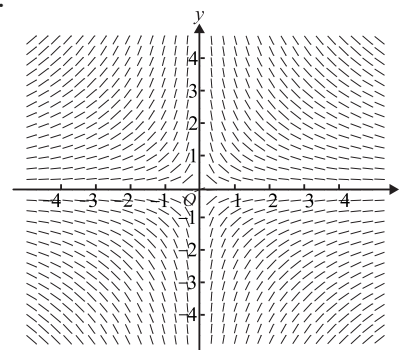
A.



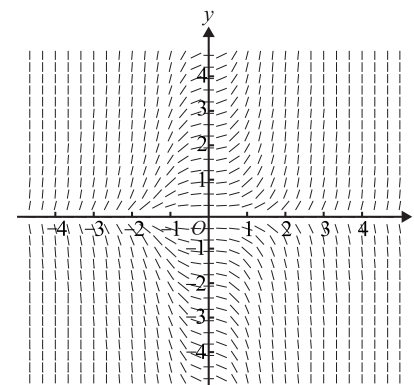
B.



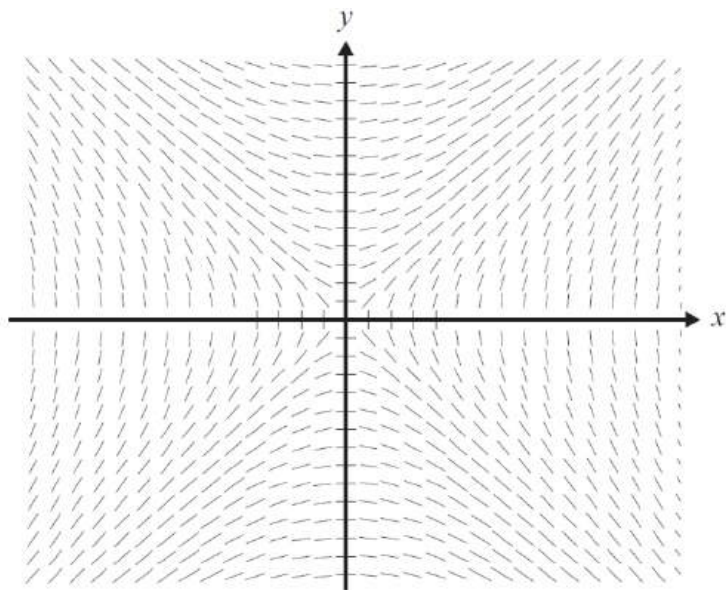
C.



D.



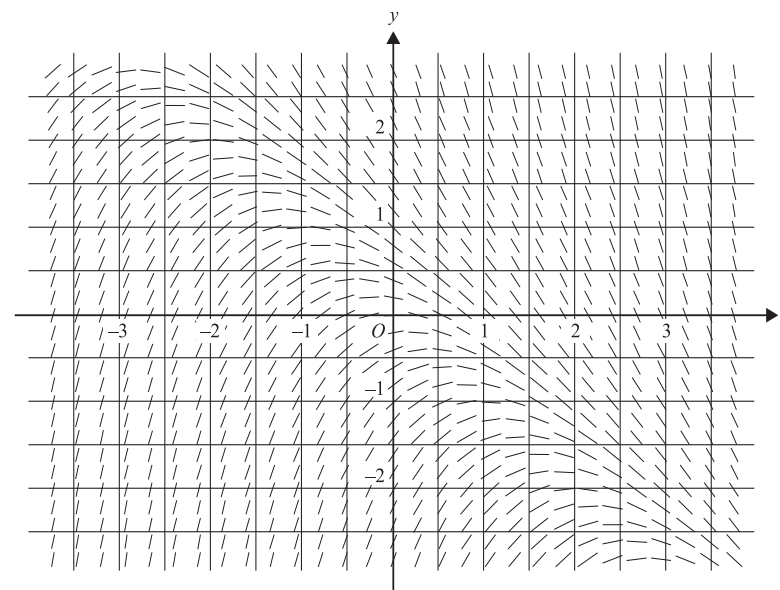
5. Calculus, EXT1 C3 2013 SPEC2 12 MC



The differential equation that best represents the above direction field is

- A. $\frac{dy}{dx} = x^2 - y^2$
- B. $\frac{dy}{dx} = y^2 - x^2$
- C. $\frac{dy}{dx} = -\frac{x}{y}$
- D. $\frac{dy}{dx} = \frac{x}{y}$

6. Calculus, EXT1 C3 2016 SPEC2 10 MC



The direction field for the differential equation $\frac{dy}{dx} + x + y = 0$ is shown above.

A solution to this differential equation that includes $(0, -1)$ could also include

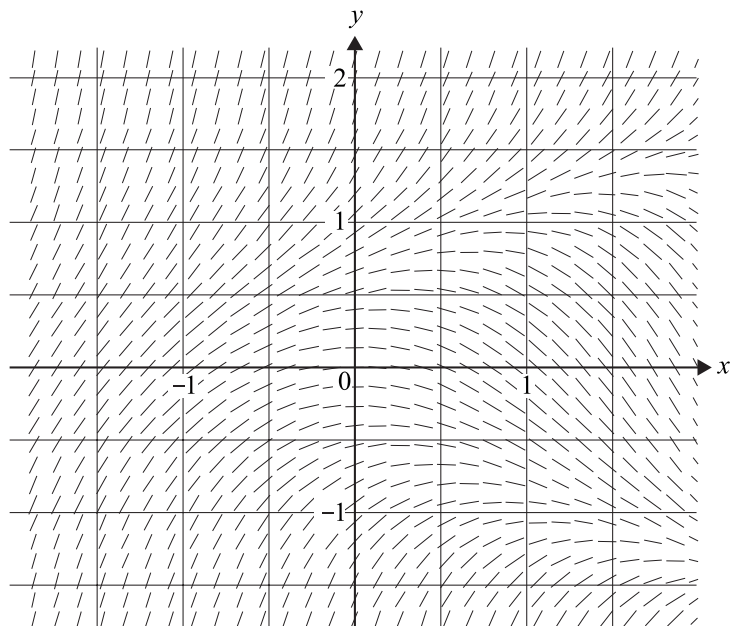
- A. $(3, -1)$
- B. $(3.5, -2.5)$
- C. $(-1.5, -2)$
- D. $(2.5, -1)$

7. Calculus, EXT1 C3 2017 SPEC2-N 10 MC

A solution to the differential equation $\frac{dy}{dx} = \frac{\cos(x+y) - \cos(x-y)}{e^{x+y}}$ can be obtained from

- A. $\int \frac{e^y}{\sin(y)} dy = - \int \frac{2\sin(x)}{e^x} dx$
- B. $\int \frac{e^y}{\cos(y)} dy = \int \frac{2}{e^x} dx$
- C. $\int \frac{e^y}{\cos(y)} dy = - \int \frac{2\cos(x)}{e^x} dx$
- D. $\int \frac{e^y}{\cos(y)} dy = \int \frac{2\sin(x)}{e^x} dx$

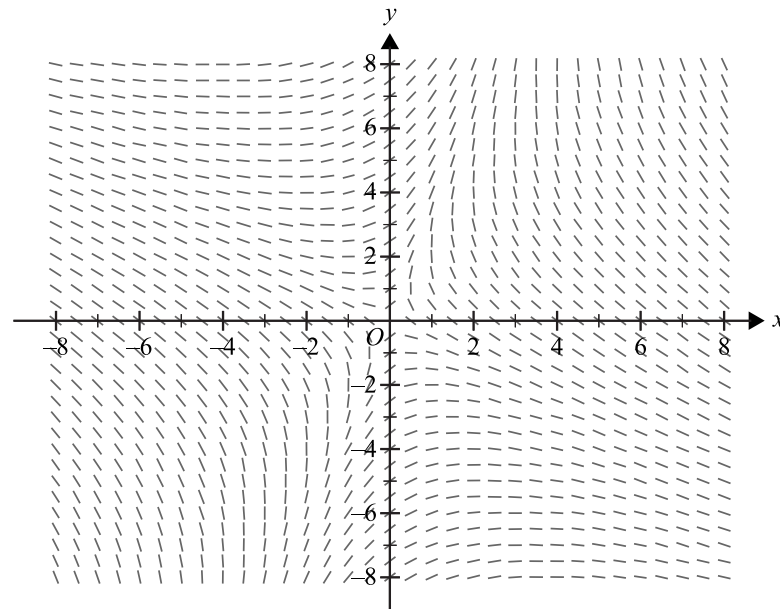
8. Calculus, EXT1 C3 2017 SPEC2 8 MC



The differential equation that best represents the direction field above is

- A. $\frac{dy}{dx} = x - y^2$
- B. $\frac{dy}{dx} = y - x$
- C. $\frac{dy}{dx} = y^2 - x^2$
- D. $\frac{dy}{dx} = y^2 - x$

10. Calculus, EXT1 C3 2018 SPEC2 10 MC



The differential equation that best represents the direction field above is

- A. $\frac{dy}{dx} = \frac{2x + y}{y - 2x}$
- B. $\frac{dy}{dx} = \frac{x + 2y}{2x - y}$
- C. $\frac{dy}{dx} = \frac{2x - y}{x + 2y}$
- D. $\frac{dy}{dx} = \frac{x - 2y}{y - 2x}$

9. Calculus, EXT1 C3 2017 SPEC2 9 MC

The gradient of the tangent to a curve at any point $P(x, y)$ is half the gradient of the line segment joining P and the point $Q(-1, 1)$.

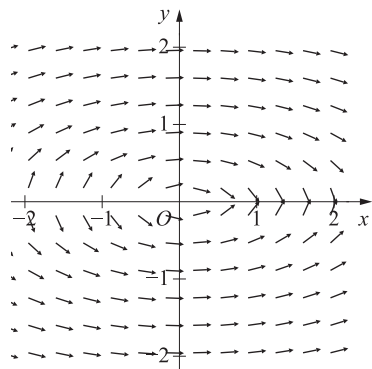
The coordinates of points on the curve satisfy the differential equation

- A. $\frac{dy}{dx} = \frac{y + 1}{2(x - 1)}$
- B. $\frac{dy}{dx} = \frac{2(y - 1)}{x + 1}$
- C. $\frac{dy}{dx} = \frac{x - 1}{2(y + 1)}$
- D. $\frac{dy}{dx} = \frac{y - 1}{2(x + 1)}$

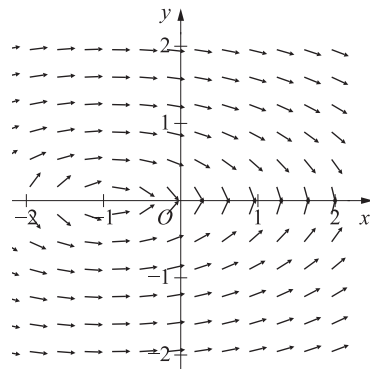
11. Calculus, EXT1 C3 2020 HSC 7 MC

Which of the following best represents the direction field for the differential equation $\frac{dy}{dx} = -\frac{x}{4y}$?

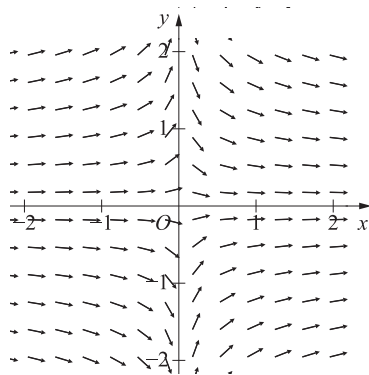
A.



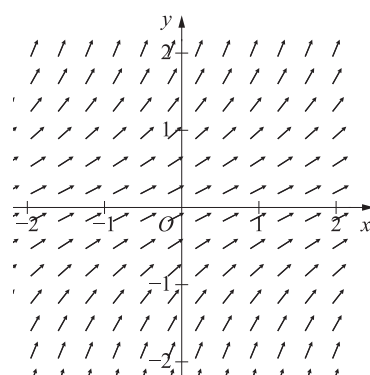
B.



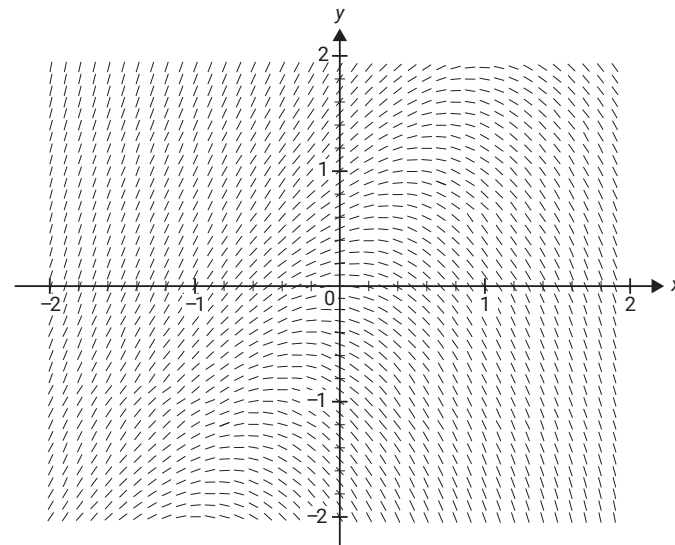
C.



D.



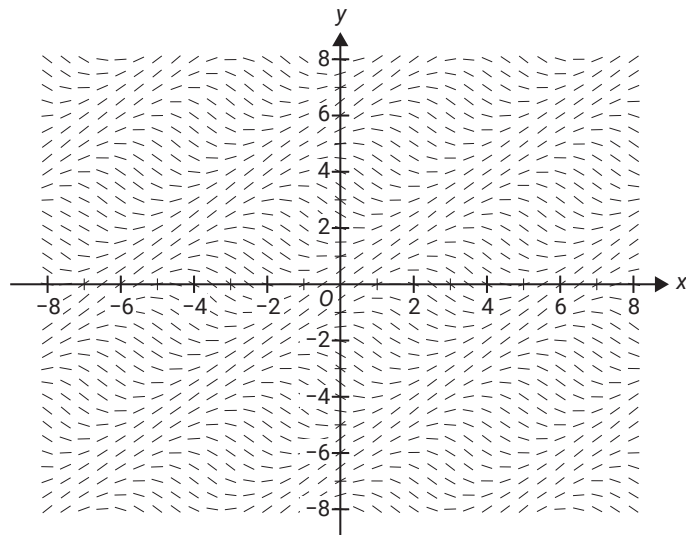
12. Calculus, EXT1 C3 2021 SPEC2 10 MC



The differential equation that has the diagram above as its direction field is

- A. $\frac{dy}{dx} = y + 2x$
- B. $\frac{dy}{dx} = 2x - y$
- C. $\frac{dy}{dx} = x + 2y$
- D. $\frac{dy}{dx} = y - 2x$

13. Calculus, EXT1 C3 2019 SPEC2 9 MC



The differential equation that has the diagram above as its direction field is

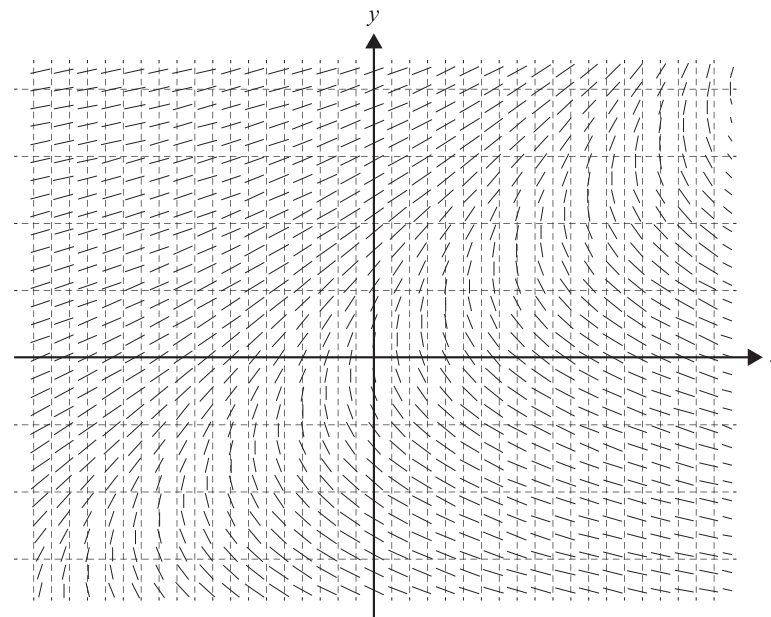
- A. $\frac{dy}{dx} = \sin(y - x)$
- B. $\frac{dy}{dx} = \cos(y - x)$
- C. $\frac{dy}{dx} = \frac{1}{\cos(y - x)}$
- D. $\frac{dy}{dx} = \frac{1}{\sin(y - x)}$

14. Calculus, EXT1 C3 2018 SPEC2 9 MC

A solution to the differential equation $\frac{dy}{dx} = \frac{2}{\sin(x + y) - \sin(x - y)}$ can be obtained from

- A. $\int 1 \, dx = \int 2\sin(y) \, dy$
- B. $\int \cos(y) \, dy = \int \operatorname{cosec}(x) \, dx$
- C. $\int \cos(x) \, dx = \int \operatorname{cosec}(y) \, dy$
- D. $\int \sec(x) \, dx = \int \sin(y) \, dy$

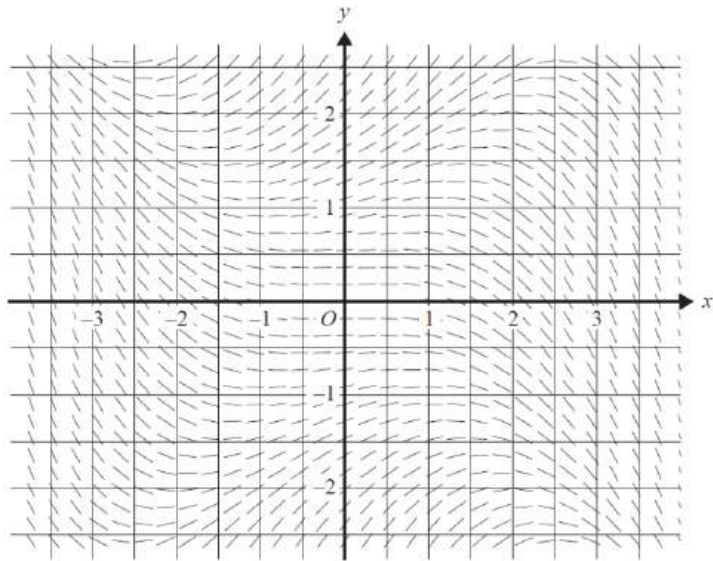
15. Calculus, EXT1 C3 2014 SPEC2 14 MC



The differential equation that is best represented by the above direction field is

- A. $\frac{dy}{dx} = \frac{1}{x - y}$
- B. $\frac{dy}{dx} = y - x$
- C. $\frac{dy}{dx} = \frac{1}{y - x}$
- D. $\frac{dy}{dx} = x - y$

16. Calculus, EXT1 C3 2015 SPEC2 13 MC



The direction field for a certain differential equation is shown above.

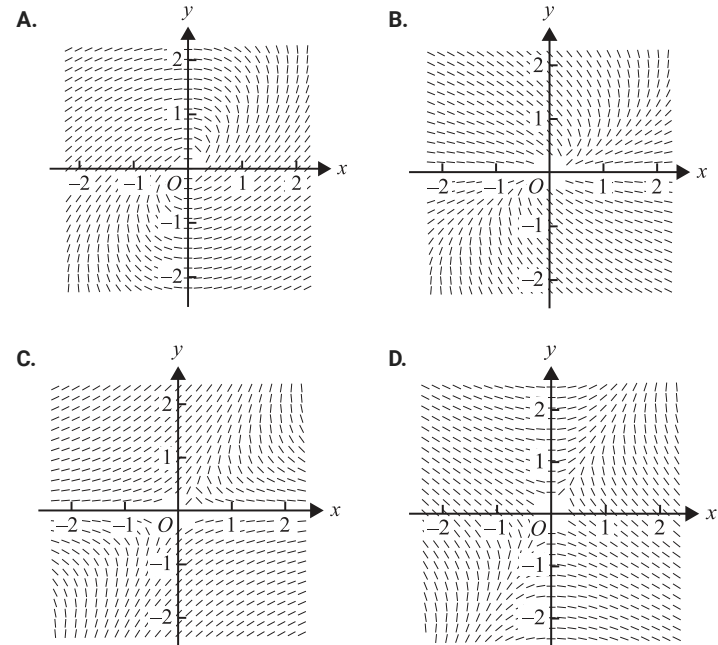
The solution curve to the differential equation that passes through the point $(-2.5, 1.5)$ could also pass through

- A. $(0, 2)$
- B. $(1, 2)$
- C. $(3, 1)$
- D. $(3, -0.5)$

17. Calculus, EXT1 C3 2020 SPEC2 9 MC

$P(x, y)$ is a point on a curve. The x -intercept of a tangent to point $P(x, y)$ is equal to the y -value at P .

Which one of the following slope fields best represents this curve?

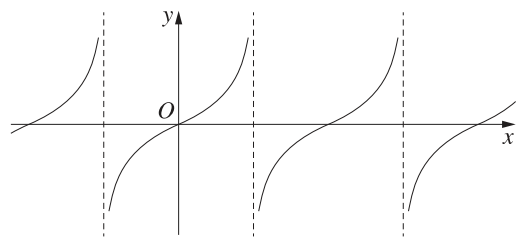


18. Calculus, EXT1 C3 2022 HSC 10 MC

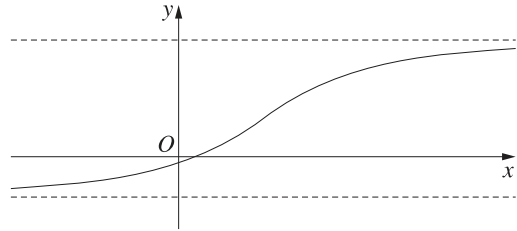
Which of the following could be the graph of a solution to the differential equation

$\frac{dy}{dx} = \sin y + 1$?

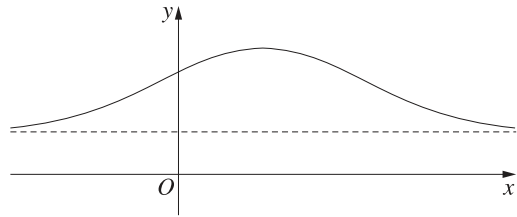
A.



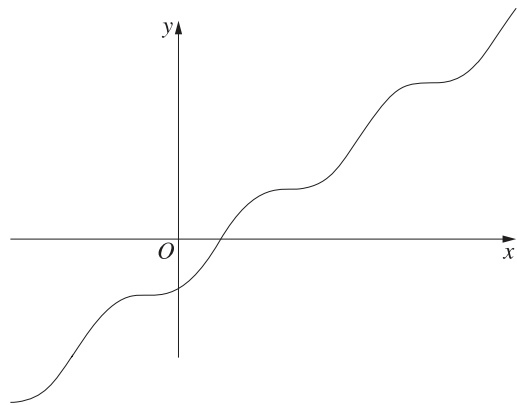
B.



C.



D.



19. Calculus, EXT1 C3 2015 SPEC2 12

Find y given $\frac{dy}{dx} = 1 - \frac{y}{3}$ and $y = 4$ when $x = 2$. (2 marks)

20. Calculus, EXT1 C3 2020 HSC 11e

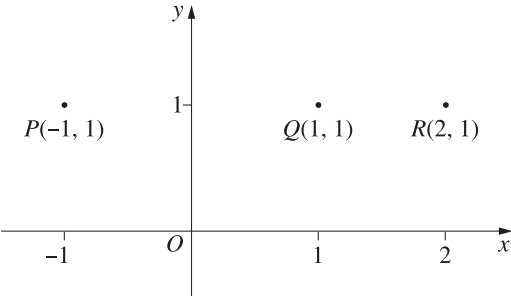
Solve $\frac{dy}{dx} = e^{2y}$, finding x as a function of y . (2 marks)

21. Calculus, EXT1 C3 2022 HSC 12a

A direction field is to be drawn for the differential equation

$\frac{dy}{dx} = \frac{x - 2y}{x^2 + y^2}$.

On the diagram, clearly draw the correct slopes of the direction field at the points P , Q and R . (2 marks)



22. Calculus, EXT1 C3 2024 HSC 11d

Solve the differential equation $\frac{dy}{dx} = xy$, given $y > 0$. Express your answer in the form $y = e^{f(x)}$. (2 marks)

23. Calculus, EXT1 C3 EQ-Bank 12

Solve $\frac{dy}{dx} = 2x^2 - 3x$ given that when $x = 3$, $y = -1$. (2 marks)

24. Calculus, EXT1 C3 2017 SPEC1-N 7

Let $\frac{dy}{dx} = (4 - y)^2$.

Express y in terms of x , where $y(0) = 3$. (3 marks)

25. Calculus, EXT1 C3 2022 SPEC1 2

Solve the differential equation $\frac{dy}{dx} = -x\sqrt{4-y^2}$ given that $y(2) = 0$. Give your answer in the form $y = f(x)$. (3 marks)

26. Calculus, EXT1 C3 EQ-Bank 13

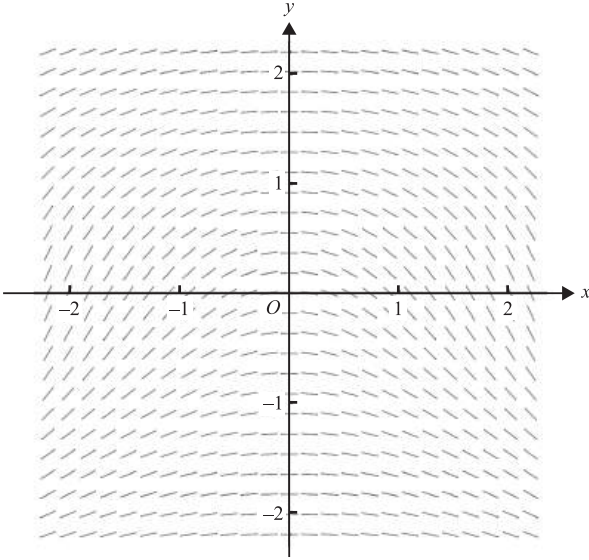
Find an expression for y in terms of x given $\frac{dy}{dx} = 4y - 3$ and when $x = -2$, $y = 1$. (3 marks)

27. Calculus, EXT1 C3 EQ-Bank 15

Find the particular solution to the differential equation $\frac{dy}{dx} = e^{2x+3y}$ that passes through the point $(0, 0)$. (3 marks)

28. Calculus, EXT1 C3 2017 SPEC1 8

A slope field representing the differential equation $\frac{dy}{dx} = -\frac{x}{1+y^2}$ is shown below.



- a. Sketch the solution curve of the differential equation corresponding to the condition $y(-1) = 1$ on the slope field above and, hence, estimate the positive value of x when $y = 0$. Give your answer correct to one decimal place. (2 marks)
- b. Solve the differential equation $\frac{dy}{dx} = \frac{-x}{1+y^2}$ with the condition $y(-1) = 1$. Express your answer in the form $ay^3 + by + cx^2 + d = 0$, where a, b, c and d are integers. (2 marks)

29. Calculus, EXT1 C3 2016 SPEC1 10

Solve the differential equation $\sqrt{2-x^2}\frac{dy}{dx} = \frac{1}{2-y}$, given that $y(1) = 0$. Express y as a function of x . (4 marks)

30. Calculus, EXT1 C3 2020 HSC 12e

Find the curve which satisfies the differential equation $\frac{dy}{dx} = -\frac{x}{y}$ and passes through the point $(1, 0)$. (3 marks)

31. Calculus, EXT1 C3 2022 HSC 14a

Find the particular solution to the differential equation $(x-2)\frac{dy}{dx} = xy$ that passes through the point $(0, 1)$. (4 marks)

32. Calculus, EXT1 C3 2024 HSC 14a

Find the domain and range of the function that is the solution to the differential equation $\frac{dy}{dx} = e^{x+y}$ and whose graph passes through the origin. (4 marks)

33. Calculus, EXT1 C3 EQ-Bank 14

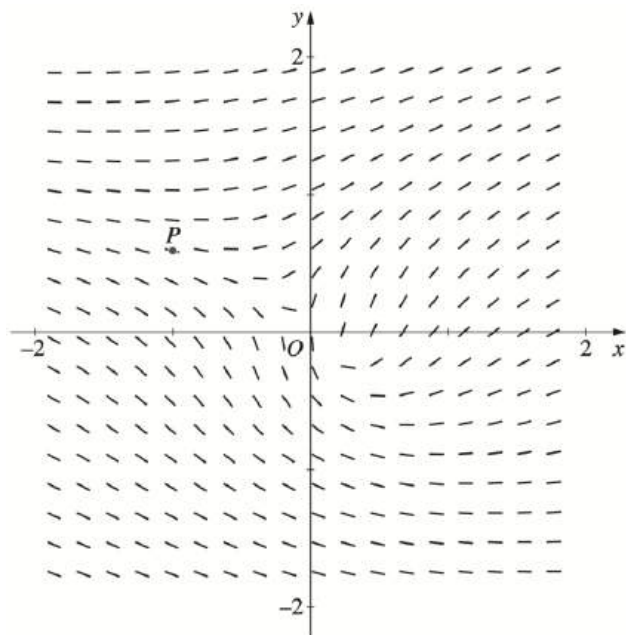
Find the particular solution to the differential equation $\frac{dy}{dx} = (2y+1)(x-3)$ that passes through the point $(2, -1)$. (4 marks)

34. Calculus, EXT1 C3 2021 HSC 12a

The direction field for a differential equation is shown below.

The graph of a particular solution to the differential equation passes through the point P .

On the graph, sketch the graph of this particular solution. (1 mark)



Worked Solutions

1. Calculus, EXT1 C3 2021 HSC 4 MC

$$\frac{dy}{dx} = \frac{x}{y}$$

$$\int y \, dy = \int x \, dx$$

$$\frac{1}{2}y^2 = \frac{1}{2}x^2 + c$$

$$y^2 = x^2 + c_1$$

$$\Rightarrow A$$

2. Calculus, EXT1 C3 2023 HSC 3 MC

Following gradients $\rightarrow$ cross y-axis slightly above 1.5

$$\Rightarrow C$$

3. Calculus, EXT1 C3 2011 SPEC2 17 MC

When $x = 0 \Rightarrow$ gradients are all positive

Eliminate B and E.

When $y = 0 \Rightarrow$ gradients are all negative

Eliminate D.

Option A will have zero gradient along $y = 2x$ (correct)

Option C will have zero gradient along $y = \frac{1}{2}x$ (incorrect)

$$\Rightarrow A$$

Almost half of all students answered incorrectly – mean mark 52%.

4. Calculus, EXT1 C3 2012 SPEC2 10 MC

$$\frac{dy}{dx} = xy$$

When $x = 0$ or $y = 0 \Rightarrow$ gradient = 0

In 1st and 3rd quartile $\Rightarrow$ gradients positive

In 2nd and 4th quartile $\Rightarrow$ gradients negative

$$\Rightarrow A$$

5. Calculus, EXT1 C3 2013 SPEC2 12 MC

By inspection:

When $x = 0 \Rightarrow \frac{dy}{dx} = 0$

When $y = 0 \Rightarrow \frac{dy}{dx} \rightarrow \infty$

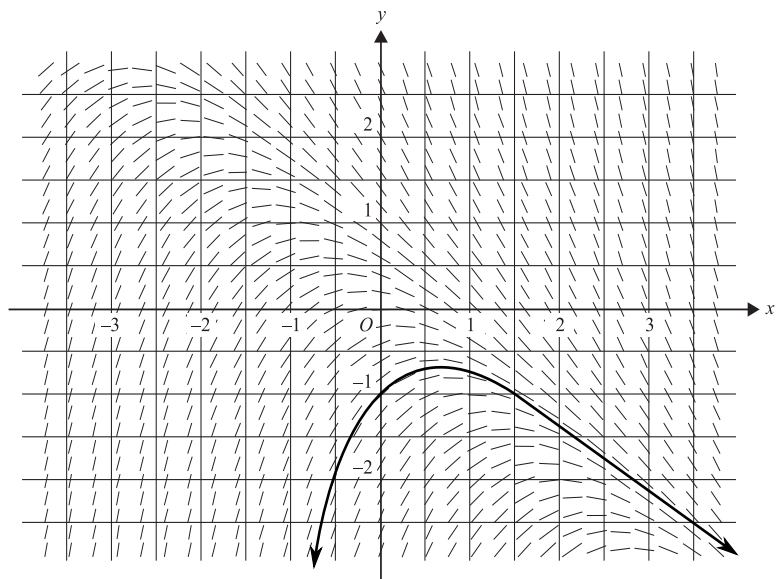
$\therefore$ Eliminate A, and B

Along $y = x \Rightarrow \frac{dy}{dx} > 0$

$\therefore$ Eliminate C

$\Rightarrow D$

6. Calculus, EXT1 C3 2016 SPEC2 10 MC



$\Rightarrow B$

7. Calculus, EXT1 C3 2017 SPEC2-N 10 MC

$$\frac{dy}{dx} = \frac{\cos(x+y) - \cos(x-y)}{e^{x+y}}$$

$$\frac{dy}{dx} = \frac{\cos(x)\cos(y) - \sin(x)\sin(y) - \cos(x)\cos(y) - \sin(x)\sin(y)}{e^x \cdot e^y}$$

$$e^y \cdot \frac{dy}{dx} = \frac{-2\sin(x)\sin(y)}{e^x}$$

$$\frac{e^y}{\sin(y)} \cdot \frac{dy}{dx} = \frac{-2\sin(x)}{e^x}$$

$$\therefore \int \frac{e^y}{\sin(y)} dy = - \int \frac{2\sin(x)}{e^x} dx$$

$\Rightarrow A$

8. Calculus, EXT1 C3 2017 SPEC2 8 MC

Use CAS to graph the direction field of each option.

$\Rightarrow D$

9. Calculus, EXT1 C3 2017 SPEC2 9 MC

$$m_{\text{tang}} = \frac{1}{2} m_{PQ}$$

$$m_{PQ} = \frac{y-1}{x-(-1)}$$

$$= \frac{y-1}{x+2}$$

$$\therefore m_{\text{tang}} = \frac{dy}{dx} = \frac{y-1}{2(x+1)}$$

$\Rightarrow D$

10. Calculus, EXT1 C3 2018 SPEC2 10 MC

When $x = 0, m = 1$

When $y = 0, m = -1$

$\Rightarrow A$

11. Calculus, EXT1 C3 2020 HSC 7 MC

At $x = 0, \frac{dy}{dx} = 0$ (horizontal)

$\Rightarrow A$

12. Calculus, EXT1 C3 2021 SPEC2 10 MC

By elimination:

At $(1, 2), m = 0$

→ Eliminate A, C

At $(0, 1), m$ is positive

→ Eliminate B

⇒ D

13. Calculus, EXT1 C3 2019 SPEC2 9 MC

By elimination:

Along line $y = x$, gradient = 1

∴ Eliminate A and D.

At $(1, 0), 0 < \text{gradient} < 1$

$$\frac{1}{\cos(-1)} > 1$$

∴ Eliminate C

⇒ B

14. Calculus, EXT1 C3 2018 SPEC2 9 MC

$$\begin{aligned} \frac{dy}{dx} &= \frac{2}{\sin(x)\cos(y) + \sin(y)\cos(x) - (\sin(x)\cos(y) - \sin(y)\cos(x))} \\ &= \frac{2}{2\sin(y)\cos(x)} \\ &= \frac{1}{\sin(y)\cos(x)} \end{aligned}$$

$$\sin(y) \cdot \frac{dy}{dx} = \sec(x)$$

$$\int \sin(y) dy = \int \sec(x) dx$$

⇒ D

15. Calculus, EXT1 C3 2014 SPEC2 14 MC

Consider quadrant 2,

$$x < 0, y > 0, m > 0 \Rightarrow \text{Eliminate A and D}$$

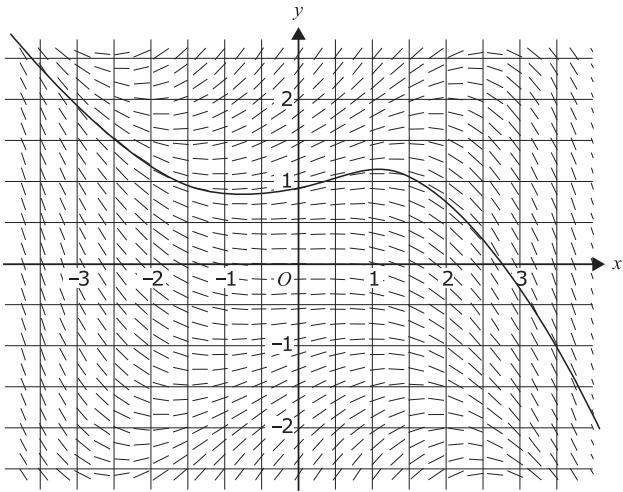
Consider vertical gradients where $m = \infty \Rightarrow \text{Eliminate B}$

⇒ C

16. Calculus, EXT1 C3 2015 SPEC2 13 MC

Draw a graph that goes through $(-2.5, 1.5)$ such that all gradient curve lines are tangential:

◆ Mean mark 47%.



$$(3, -0.5)$$

⇒ D

17. Calculus, EXT1 C3 2020 SPEC2 9 MC

The tangent to the curve passes through

$$(x, y) \text{ and } (y, 0)$$

$$\frac{dy}{dx} = \frac{0-y}{y-x} = \frac{y}{x-y}$$

When $x = 0$:

$$\frac{dy}{dx} = \frac{y}{-y} = -1$$

⇒ B

18. Calculus, EXT1 C3 2022 HSC 10 MC

One Strategy

When $\frac{dy}{dx} = 0$:

◆◆ Mean mark 27%.

$$\sin y = -1 \Rightarrow y = \frac{3\pi}{2} + 2k\pi \quad (k \in \mathbb{Z})$$

Graphically, $y = \frac{3\pi}{2} + 2k\pi$ are horizontal asymptotes.

⇒ B

19. Calculus, EXT1 C3 2015 SPEC2 12

$$\frac{dy}{dx} = \frac{3-y}{3}$$
$$\frac{dx}{dy} = \frac{3}{3-y}$$
$$x = \int \frac{3}{3-y} dy$$
$$\frac{x}{3} = -\ln|3-y| + c$$

Given $y = 4$ when $x = 2$:

$$\frac{2}{3} = -\ln|-1| + c$$
$$c = \frac{2}{3}$$
$$\frac{x}{3} = -\ln|3-y| + \frac{2}{3}$$
$$\ln|3-y| = \frac{2-x}{3}$$
$$3-y = \pm e^{\frac{2-x}{3}}$$
$$\therefore y = 3 + e^{\frac{2-x}{3}}$$

20. Calculus, EXT1 C3 2020 HSC 11e

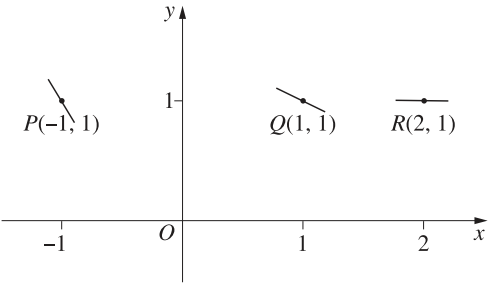
$$\frac{dy}{dx} = e^{2y}$$
$$\frac{dx}{dy} = e^{-2y}$$
$$x = \int e^{-2y} dy$$
$$\therefore x = -\frac{1}{2}e^{-2y} + c$$

21. Calculus, EXT1 C3 2022 HSC 12a

At $(-1, 1)$: $\frac{dy}{dx} = \frac{-1-2}{1+1} = -\frac{3}{2}$

At $(1, 1)$: $\frac{dy}{dx} = \frac{1-2}{1+1} = -\frac{1}{2}$

At $(2, 1)$: $\frac{dy}{dx} = \frac{2-2}{4+1} = 0$



22. Calculus, EXT1 C3 2024 HSC 11d

$$\frac{dy}{dx} = xy$$
$$\int \frac{1}{y} dy = \int x dx$$
$$\ln y = \frac{1}{2}x^2 + c$$
$$y = e^{\frac{x^2}{2} + c}$$
$$= e^{\frac{x^2}{2}} \cdot e^c$$
$$= Ae^{\frac{x^2}{2}} \text{ (where } A = e^c \text{)}$$
$$\therefore y = e^{\frac{x^2}{2}} \text{ is a solution (} A = 1 \text{)}$$

23. Calculus, EXT1 C3 EQ-Bank 12

$$\frac{dy}{dx} = 2x^2 - 3x$$

$$\int 1 \, dy = \int 2x^2 - 3x \, dx$$

$$\int 1 \, dy = \int 2x^2 - 3x \, dx$$

$$y = \frac{2}{3}x^3 - \frac{3}{2}x^2 + c$$

When $x = 3$, $y = -1$:

$$-1 = \frac{2}{3}3^3 - \frac{3}{2}3^2 + c$$

$$c = -\frac{11}{2}$$

$$\therefore y = \frac{2}{3}x^3 - \frac{3}{2}x^2 - \frac{11}{2}$$

24. Calculus, EXT1 C3 2017 SPEC1-N 7

$$\frac{dy}{dx} = (4 - y)^2$$

$$\frac{dx}{dy} = \frac{1}{(4 - y)^2}$$

$$x = \int \frac{1}{(4 - y)^2} \, dy$$

$$= \int (4 - y)^{-2} \, dy$$

$$= (-1)(-1)(4 - y)^{-1} + c$$

$$= \frac{1}{4 - y} + c$$

When $x = 0$, $y = 3$:

$$0 = \frac{1}{4 - 3} + c$$

$$\therefore c = -1$$

$$x = \frac{1}{4 - y} - 1$$

$$x + 1 = \frac{1}{4 - y}$$

$$\frac{1}{x + 1} = 4 - y$$

$$\therefore y = 4 - \frac{1}{x + 1}$$

25. Calculus, EXT1 C3 2022 SPEC1 2

$$\int \frac{dy}{\sqrt{4 - y^2}} = \int -x \, dx$$

$$\sin^{-1}\left(\frac{y}{2}\right) = -\frac{1}{2}x^2 + c$$

$$y(2) = 0 \Rightarrow c = 2$$

$$\frac{y}{2} = \sin\left(-\frac{1}{2}x^2 + 2\right)$$

$$y = 2\sin\left(-\frac{1}{2}x^2 + 2\right)$$

26. Calculus, EXT1 C3 EQ-Bank 13

$$\frac{dy}{dx} = 4y - 3$$

$$\frac{dy}{4y - 3} = 1 \, dx$$

$$\int \frac{1}{4y - 3} \, dy = \int 1 \, dx$$

$$\frac{1}{4} \ln|4y - 3| = x + c$$

When $y = 1, x = -2$:

$$\frac{1}{4} \ln(4 - 3) = -2 + c$$

$$c = 2$$

$$\frac{1}{4} \ln|4y - 3| = x + 2$$

$$\ln|4y - 3| = 4(x + 2)$$

$$4y - 3 = \pm e^{4(x+2)}$$

$$4y - 3 = e^{4(x+2)}, \quad (\text{since } y(-2) = 1)$$

$$4y = e^{4(x+2)} + 3$$

$$y = \frac{e^{4(x+2)} + 3}{4}$$

27. Calculus, EXT1 C3 EQ-Bank 15

$$\frac{dy}{dx} = e^{2x+3y}$$

$$\frac{dy}{dx} = e^{2x} \cdot e^{3y}$$

$$e^{-3y} \, dy = e^{2x} \, dx$$

$$\int e^{-3y} \, dy = \int e^{2x} \, dx$$

$$-\frac{1}{3}e^{-3y} = \frac{1}{2}e^{2x} + c$$

Passes through $(0, 0)$:

$$-\frac{1}{3}e^0 = \frac{1}{2}e^0 + c$$

$$c = -\frac{5}{6}$$

$$-\frac{1}{3}e^{-3y} = \frac{1}{2}e^{2x} - \frac{5}{6}$$

$$2e^{-3y} = 5 - 3e^{2x}$$

$$e^{-3y} = \frac{5 - 3e^{2x}}{2}$$

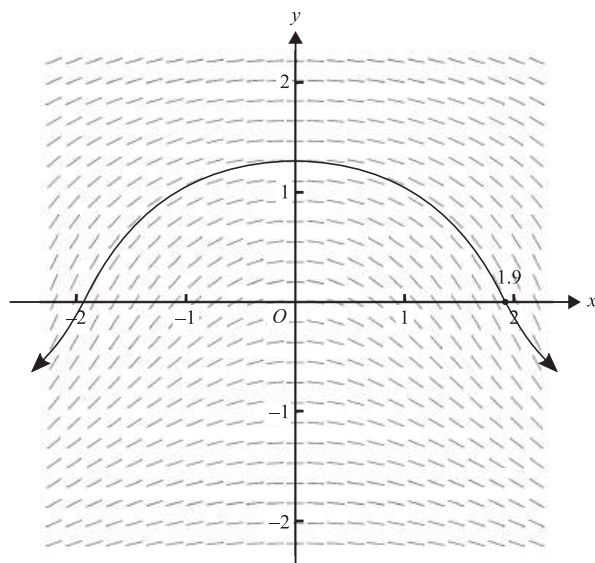
$$\ln(e^{-3y}) = \ln\left(\frac{5 - 3e^{2x}}{2}\right)$$

$$-3y = \ln\left(\frac{5 - 3e^{2x}}{2}\right)$$

$$y = -\frac{1}{3} \ln\left(\frac{5 - 3e^{2x}}{2}\right)$$

28. Calculus, EXT1 C3 2017 SPEC1 8

a.



♦♦ Mean mark part (a) 32%.
MARKER'S COMMENT: Solution curve should follow slope ticks and not cross them.

b. $(1 + y^2) \frac{dy}{dx} = -x$

$$\int 1 + y^2 dy = - \int x dx$$

$$y + \frac{y^3}{3} = -\frac{x^2}{2} + C, C \in \mathbb{R}$$

Substituting $(-1, 1)$:

$$1 + \frac{1^3}{3} = -\frac{(-1)^2}{2} + C$$

$$1 + \frac{1}{3} = -\frac{1}{2} + C$$

$$\therefore C = \frac{11}{6}$$

$$y + \frac{1}{3}y^3 = -\frac{1}{2}x^2 + \frac{11}{6}$$

$$6y + 2y^3 = -3x^2 + 11$$

$$\therefore 2y^3 + 6y + 3x^2 - 11 = 0$$

29. Calculus, EXT1 C3 2016 SPEC1 10

$$\sqrt{2 - x^2} \cdot \frac{dy}{dx} = \frac{1}{2 - y}$$

$$(2 - y) \cdot \frac{dy}{dx} = \frac{1}{\sqrt{2 - x^2}}$$

$$\int 2 - y dy = \int \frac{1}{\sqrt{2 - x^2}} dx$$

$$2y - \frac{y^2}{2} = \sin^{-1}\left(\frac{x}{\sqrt{2}}\right) + c$$

Given $y(1) = 0$:

$$0 = \sin^{-1}\left(\frac{1}{\sqrt{2}}\right) + c$$

$$c = -\frac{\pi}{4}$$

$$2y - \frac{y^2}{2} = \sin^{-1}\left(\frac{x}{\sqrt{2}}\right) - \frac{\pi}{4}$$

$$y^2 - 4y = -2\sin^{-1}\left(\frac{x}{\sqrt{2}}\right) + \frac{\pi}{2}$$

$$(y - 2)^2 - 4 = -2\sin^{-1}\left(\frac{x}{\sqrt{2}}\right) + \frac{\pi}{2}$$

$$(y - 2)^2 = 4 + \frac{\pi}{2} - 2\sin^{-1}\left(\frac{x}{\sqrt{2}}\right)$$

$$(y - 2) = \pm \sqrt{4 + \frac{\pi}{2} - 2\sin^{-1}\left(\frac{x}{\sqrt{2}}\right)}$$

$$y = 2 \pm \sqrt{4 + \frac{\pi}{2} - 2\sin^{-1}\left(\frac{x}{\sqrt{2}}\right)}$$

Given $y = 0$ when $x = 1$:

$$\therefore y = 2 - \sqrt{4 + \frac{\pi}{2} - 2\sin^{-1}\left(\frac{x}{\sqrt{2}}\right)}$$

♦ Mean mark 46%.

30. Calculus, EXT1 C3 2020 HSC 12e

$$\frac{dy}{dx} = -\frac{x}{y}$$

$$\int y \, dy = -\int x \, dx$$

$$\frac{y^2}{2} = -\frac{x^2}{2} + c$$

Curve passes through (1, 0):

$$0 = -\frac{1}{2} + c$$

$$c = \frac{1}{2}$$

$$\frac{y^2}{2} = -\frac{x^2}{2} + \frac{1}{2}$$

$$y^2 = -x^2 + 1$$

$$\therefore x^2 + y^2 = 1$$

COMMENT: Note the answer requires a curve equation, not a function.

31. Calculus, EXT1 C3 2022 HSC 14a

$$(x-2)\frac{dy}{dx} = xy$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = \frac{x}{x-2}$$

$$\int \frac{1}{y} \, dy = \int \frac{x}{x-2} \, dx$$

$$\ln|y| = \int \frac{x-2}{x-2} + \frac{2}{x-2} \, dx$$

$$= \int 1 + \frac{2}{x-2} \, dx$$

$$= x + 2\ln|x-2| + c$$

Passes through (0,1):

$$\ln 1 = 0 + 2\ln|-2| + c$$

$$c = -2\ln 2$$

$$\ln|y| = x + 2\ln|x-2| - 2\ln 2$$

$$= \ln e^x + \ln(x-2)^2 - \ln 2^2$$

$$= \ln \left(e^x \frac{(x-2)^2}{4} \right)$$

$$|y| = \frac{e^x (x-2)^2}{4}$$

$$\therefore y = \frac{e^x (x-2)^2}{4} \quad (e^x > 0, \quad (x-2)^2 > 0)$$

◆ Mean mark 43%.

32. Calculus, EXT1 C3 2024 HSC 14a

$$\frac{dy}{dx} = e^{x+y} = e^x \cdot e^y$$

$$\int e^{-y} dy = \int e^x dx$$

$$-e^{-y} = e^x + c$$

Passes through $(0, 0)$:

$$-e^0 = e^0 + c \Rightarrow c = -2$$

$$-e^{-y} = e^x - 2$$

$$e^{-y} = 2 - e^x$$

$$-y = \ln(2 - e^x)$$

$$y = -\ln(2 - e^x)$$

Since $2 - e^x > 0 \Rightarrow e^x < 2$

$\Rightarrow$ Domain: $x < \ln 2$

Since $e^x > 0 \Rightarrow 2 - e^x < 2$

$\Rightarrow$ Range: $y > -\ln 2$

◆ Mean mark 52%.

33. Calculus, EXT1 C3 EQ-Bank 14

$$\frac{dy}{dx} = (2y + 1)(x - 3)$$

$$\frac{dy}{2y + 1} = (x - 3) dx$$

$$\int \frac{1}{2y + 1} dy = \int (x - 3) dx$$

$$\frac{1}{2} \ln|2y + 1| = \frac{x^2}{2} - 3x + c$$

Passes through $(2, -1)$:

$$\frac{1}{2} \ln|-1| = 2 - 6 + c$$

$$c = 4$$

$$\frac{1}{2} \ln|2y + 1| = \frac{x^2}{2} - 3x + 4$$

$$\ln|2y + 1| = x^2 - 6x + 8$$

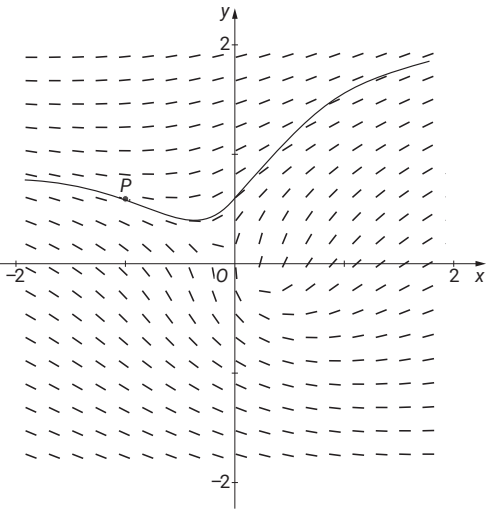
$$\ln|2y + 1| = (x - 4)(x - 2)$$

$$2y + 1 = \pm e^{(x-2)(x-4)}$$

$$2y = -e^{(x-2)(x-4)} - 1, \text{ (passes through } (2, -1))$$

$$y = -\frac{1}{2} \left(e^{(x-2)(x-4)} + 1 \right)$$

34. Calculus, EXT1 C3 2021 HSC 12a



◆◆◆ Mean mark 18%!
MARKER COMMENT: A solution curve does not cross any tangent line.

